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Diagnostic Plots to Reveal Functional Form for Covariates in Multiplicative Intensity Models

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SUMMARY

We show how plots based on the residuals from a proportional hazards model may be used to reveal the correct functional form for covariates in the model. A smoothed plot of the martingale residuals was suggested for this purpose by Therneau, Grambsch, and Fleming (1990, *Biometrika* **77**, 147–160); however, its consistency required that the covariates be independent. They also noted that the plot could be biased for large covariate effects. We introduce two refinements which overcome these difficulties. The first is based on a ratio of scatter plot smooths, where the numerator is the smooth of the observed count plotted against the covariate, and the denominator is a smooth of the expected count. This is related to the Arjas goodness-of-fit plot (1988, *Journal of the American Statistical Association* **83**, 204–212). The second technique smooths the martingale residuals divided by the expected count, using expected count as a weight. This latter approach is related to a GLM partial residual plot, as well as to the iterative methods of Hastie and Tibshirani (1990, *Biometrics* **46**, 1005–1016) and Gentleman and Crowley (1991, *Biometrics* **47**, 1283–1296). Applications to survival data sets are given.

1. Introduction

Suppose we have data from a multiplicative intensity model (Fleming and Harrington (1991), Definition 4.2.1) consisting of n independent triples $\{N_i(t), Y_i(t), \mathbf{Z}_i, i = 1, 2, \dots, n; 0 \leq t \leq \tau\}$, where $N_i(t)$ is a counting process counting the number of observed events up to time t , $Y_i(t)$ is a 0–1 predictable process indicating whether subject i is at a risk for an observed event at time t , and $\mathbf{Z}_i$ is a p vector of baseline covariates (not time-dependent) for subject i . We assume that $P\{Y(t) > 0\} > 0$ on $[0, \tau]$. The intensity process $l_i(t)$ is modeled as $Y_i(t)\exp(\beta'\mathbf{Z}_i)\lambda_0(t)$, where β is a vector of unknown regression coefficients and $\Lambda_0 = \int \lambda_0$ is the baseline cumulative hazard function. The random process $\int_0^t l_i(s) ds$, to be denoted $E_i(t)$, is called the compensator for $N_i(t)$ and $E\{N_i(t)|\mathbf{Z}\} = E\{E_i(t)|\mathbf{Z}\}$.

Thus, we assume a proportional hazards structure with a log-linear model for the covariates. The investigator wants to discover their correct functional form. Perhaps one of the variables, Z_j , should be replaced by Z_j^2 , $\ln Z_j$, $I_{\{Z_j > c\}}$ or some other transform. This issue was examined by Therneau, Grambsch, and Fleming (1990) who suggested smoothed residual plots. In this paper, we suggest two refinements which offer substantial improvements. Section 2 provides the theoretical development and discusses the relationships to Arjas' (1988) martingale process plots, to GLM partial residuals and to the penalized likelihood approach of Hastie and Tibshirani (1990a, b). Section 3 shows the behavior of the proposed techniques for a Cox model, using simulated data. Section 4 gives examples of data applications, and Section 5 concludes with a discussion.

Key words: Augmented partial residuals; Cox model; Partial residuals; Proportional hazards model.

2. Theoretical Development

2.1 Ratio Estimator

To motivate our procedure, consider the instructive but unrealistic special case where the investigator knows the intensity process except for the functional form for one particular covariate, denoted X for clarity in this section. Thus, in the hazard $h(X)e^{\beta'Z}\lambda_0(t)$, the investigator knows that the functional form for the covariates $\mathbf{Z}$ is $\beta'\mathbf{Z}$ as well as the values for β and $\lambda_0(t)$ when X is entered correctly into the model, but does not know the functional form for X . To estimate $h(X)$ note that

$$E\{N(t)|X\} = h(X)E\left\{\int_0^t Y(s)e^{\beta'Z} d\Lambda_0(s)|X\right\}, \quad 0 \leq t \leq \tau, \quad (1)$$

where the right-hand expectation is over the conditional joint distribution of $\mathbf{Z}$ and Y given X . Using data over the entire interval $[0, \tau]$, we have

$$\begin{aligned} h(X) &= \frac{E\{N(\tau)|X\}}{E\left\{\int_0^\tau Y(s)e^{\beta'Z} d\Lambda_0(s)|X\right\}} \\ &= \frac{E(\text{observed count}|X)}{E(\text{expected count}|X)}. \end{aligned}$$

In the case of replicate X values, the numerator and denominator can be estimated by simple averages. In the case of unique X values, the numerator can be estimated by a smooth applied to the scatter plot of the observed count against X . Similarly, the denominator can be estimated by smoothing the scatter plot of the expected counts. A plot of the ratio, estimated numerator over estimated denominator, will suggest the functional form for $h(X)$.

Consider the more realistic case where only the functional form for the $\mathbf{Z}$ covariates is known. To mimic the above procedure, the investigator must estimate the values that β and $\Lambda_0(t)$ would take if X were entered correctly into the model. This requires an initial guess for the functional form for X . A log-linear form is often adequate and we use that form for illustration. The investigator fits the model

$$Y(s)\exp(\beta'Z)\exp(\gamma X)d\Lambda_0(s).$$

Let

$$\hat{E}_i = \exp(\hat{\beta}'Z_i) \int_0^\tau Y_i(t)\exp(\hat{\gamma}X_i) d\hat{\Lambda}_0(t)$$

denote the estimated compensator for the i th individual. Let $N_i = N_i(\tau)$. To reveal $f(X) = \log h(X)$, one plots $\log\left\{\frac{\text{smooth}(N)}{\text{smooth}(\hat{E})}\right\} + \hat{\gamma}X$ against X .

Additional motivation for this procedure can be obtained through asymptotic arguments. The sampling framework can involve either independent identically distributed covariate vectors, or a deterministic sequence of vectors, fixed by experimental design. In the former case, the asymptotic framework means a sequence of experiments of increasing sample size n , sampling independent, identically distributed triples $\{N_i(t), Y_i(t), Z_i; i = 1, \dots, n; 0 \leq t \leq \tau\}$. In the latter case, one samples independent pairs $\{N_i(t), Y_i(t), i = 1, \dots, n; 0 \leq t \leq \tau\}$ from the conditional distribution, conditioning on $\{Z_i; i = 1, \dots, n\}$. Andersen and Gill (1982) discuss regularity conditions and the interrelationships between the two approaches.

Suppose the initial guess is correct. Under standard regularity conditions, $(\hat{\beta}, \hat{\gamma})$ converges in probability to (β, γ) and $\hat{\Lambda}_0$ converges to Λ_0 uniformly in probability over t in $[0, \tau]$. As a result, standard arguments show that

$$\hat{E} = \int_0^\tau \{Y(t)\exp(\beta'Z + \gamma X) d\Lambda_0(t)\} + o_p(1).$$

If the initial guess is wrong, then, according to theory for maximum likelihood estimation with misspecified models (White (1982) and Bickel et al. (1993), Theorem 7.4.1 c, p. 325), there exists values γ^* and β^* such that $\hat{\gamma} \rightarrow \gamma^*$ and $\hat{\beta} \rightarrow \beta^*$ under regularity conditions. In turn, by the Appendix,

$$\hat{E} = E^* + o_p(1), \quad (2)$$

where $E^* = \int_0^\tau Y(t) \exp(\beta^{*'} \mathbf{Z} + \gamma^* X) \bar{h}(t) \lambda_o(t) dt$, and where $\bar{h}$ is defined in the Appendix.

It can be shown that

$$E(E^*|X) = \bar{h} \exp(\gamma^* X) E \left\{ \int_0^\tau Y(t) \exp(\beta' \mathbf{Z}) d\Lambda_0(t) | X \right\} + e(X), \quad (3)$$

where $\bar{h}$ is constant (1 if the guess were correct) and $e(X)$ is a remainder function which will often be small relative to the leading term, particularly when the initial guess is a good approximation to the true functional form. A formal proof giving regularity conditions under which $e(X)$ is asymptotically negligible appears to be difficult and has not yet been obtained.

Equations (1)–(3) motivate the approximation

$$\begin{aligned} \frac{E(N|X)}{E(\hat{E}|X)} &\approx \frac{h(X) E \left\{ \int_0^\tau Y(t) \exp(\beta' \mathbf{Z}) d\Lambda_0(t) | X \right\}}{\bar{h} \exp(\gamma^* X) E \left\{ \int_0^\tau Y(t) \exp(\beta' \mathbf{Z}) d\Lambda_0(t) | X \right\} + e(X)} \\ &\approx \bar{h}^{-1} h(X) \exp(-\gamma^* X), \end{aligned}$$

when $e(X)$ is small. Thus

$$\log\{E(N|X)\} - \log\{E(\hat{E}|X)\} \approx \log\{h(X)\} - \gamma^* X - \log \bar{h}.$$

A plot of consistent estimators of the left-hand side against X will suggest how the true functional form deviates from the postulated log-linear model. Adding a consistent estimator of $\gamma^* X$ to the plot will suggest the functional form itself.

When the covariate of interest X is discrete with only a few levels, our plot is related to the Arjas martingale residual plot (Arjas (1988), Andersen et al. (1993), §VII 3.4., Aalen (1993)). For this situation, we introduce double subscripts. Let $N_{ij}(t)$ be the counting process for the j th individual, $j = 1, \dots, n_i$, with covariate value X_i , $i = 1, \dots, I$, with an analogous definition for $Y_{ij}(t)$ and $\mathbf{Z}_{ij}$. The asymptotic framework requires that $\lim_{n \rightarrow \infty} n_i/n = \xi_i$, where $0 < \xi_i$ for $i = 1, \dots, I$ and $\sum \xi_i = 1$. Let $N_{i.}(t) = \sum_{j=1}^{n_i} N_{ij}(t)$ and $\hat{E}_{i.}(t) = \sum_{j=1}^{n_i} \hat{E}_{ij}(t)$ be the sums of the observed and estimated expected counts within stratum i . The proposed diagnostic plots $\log\{N_{i.}(\tau)/\hat{E}_{i.}(\tau)\}$ against X_i . Let $\bar{E}_{i.}(t) = \sum_{j=1}^{n_i} \bar{E}_{ij}(t)$ denote the estimated expected count for a model omitting X from the set of fitted covariates. The Arjas method plots $N_{i.}(t)$ on the abscissa and $\bar{E}_{i.}(t)$ on the ordinate for all event times, forming connected lines, one for each stratum. When the postulated model is correct, the lines follow the 45° line. Departures from this pattern can be used to diagnose various model failures, including nonproportional hazards and omitted covariates. Our focus differs from Arjas' in that we assume that there is a covariate transformation for which proportionality of hazards holds. With an adaption of the Arjas plot obtained by replacing $\bar{E}_{i.}(t)$ with $\hat{E}_{i.}(t)$, our proposed plot would be obtained by using the points from the extreme right-hand edge of Arjas' plot and by plotting the log of the abscissa to ordinate ratio against the X values.

Now, consider the more typical case in which X is continuous and replicate values are rare, occurring only due to rounding. One cannot estimate $E(N|X)$ and $E(\hat{E}|X)$ simply by averaging. There are two alternatives. First, one may form strata by grouping adjacent values of X and then averaging within the strata, as above. Second, one may estimate $E(N|X)$ by smoothing the scatterplot of N versus X and estimate $E(\hat{E}|X)$ by smoothing the scatterplot of $\hat{E}$ versus X . Either approach introduces additional complications, because one must choose the number and nature of the groupings for the first alternative and the type and bandwidth of the smooth for the second. Both choices introduce bias. The smooth can have less bias at the price of increased variance and we chose that approach. The basic diagnostic plot becomes a plot of $\log \left\{ \frac{\text{smooth}(N)}{\text{smooth}(\hat{E})} \right\} + \hat{\gamma} X$ versus X .

Examining a ratio of scatterplot smooths is not a new idea. Hjort (1992) suggested the ratio of a smooth of the observed count of failures plotted against time to a smooth of the exposure estimate

plotted against time as a hazard estimator and showed it had good properties. Our approach is even more similar to that suggested in Therneau et al., (1990), which we will denote by TGF. A little algebra applied to equation (7) in their paper shows that they recommended plotting the log of the ratio of a smoothed observed count to a smoothed expected count, but the expected count was based on a model *omitting* X . The asymptotic properties of this approach can be derived as in the Appendix, but with $\gamma^* X_i \equiv 0$. By omitting X , the remainder term in equation (1) will be at least as large as if X were entered as a log-linear term and, if the log of the hazard has an appreciable monotonic component in X , the remainder term will be substantially larger. The bias of the TGF procedure will exceed that of the proposed procedure.

2.2 Partial Residuals

The second form of residual builds on the close relationship between counting process models and Poisson regression. To estimate $f(X)$, recall that $f(X) = \log h(X)$ and replace

$$\log(N_i/\hat{E}_i) + \hat{\gamma}X_i = \log\left[1 + \frac{N_i - \hat{E}_i}{\hat{E}_i}\right] + \hat{\gamma}X_i$$

with its one-term Taylor expansion

$$\frac{N_i - \hat{E}_i}{\hat{E}_i} + \hat{\gamma}X_i = \frac{\hat{M}_i}{\hat{E}_i} + \hat{\gamma}X_i,$$

where $\hat{M}_i$ is the martingale residual (Barlow and Prentice, 1988; Therneau et al., 1990). A plot of $(\hat{M}_i/\hat{E}_i) + \hat{\gamma}X_i$ against X_i is formally identical to the GLM partial residual plot for Poisson regression discussed by McCullagh and Nelder (1989, p. 402), who recommend smoothing the plot to discover functional form for covariates. In the case of counting process data, it is important to use a weighted scatterplot smooth with $\hat{E}_i$ values as weights to equalize variance and reduce bias. When the postulated functional form is correct, $\text{var}(M_i|X_i, \mathbf{Z}_i) = E(E_i|X_i, \mathbf{Z}_i)$, so this procedure essentially corresponds to weighting inversely as the estimated variance. Consider again the situation where X is discrete, taking values X_i , $i = 1, \dots, I$, with n_i replicates for each one. The weighted average of $\hat{M}_{ij}/\hat{E}_{ij}$ with weights $\hat{E}_{ij}$ is $(N_i/\hat{E}_{i.}) - 1$. When the postulated functional form is correct, the asymptotic distribution of $n_i^{1/2} \{N_i/\hat{E}_{i.} - 1\}$ is normal with mean 0 and the same variance as $n_i^{1/2} \log(N_i/\hat{E}_{i.})$. If one averages without weighting, the diagnostic statistic becomes $n_i^{-1} \sum_j N_{ij}/\hat{E}_{ij} - 1$. The asymptotic distribution may not have mean 0 even when the functional form is correct.

In general, $(N_i/\hat{E}_{i.})$ converges in probability to $E(N|X_i)/E(\hat{E}|X_i)$. From equation (3), this limit is approximately $\tilde{h}^{-1} \exp\{f(X) - \gamma^* X\}$, when the initial guess is a reasonable approximation to the true form. A one-term Taylor's expansion gives $f(X) - \gamma^* X - \log(\tilde{h})$. Thus, a plot of $\hat{M}_i/\hat{E}_{i.} + \hat{\gamma}X_i$ versus X_i can suggest a one-term Taylor's expansion approximation to $f(X)$ in large samples. When replicates are not available, weighted smoothing replaces averaging.

The weighted smoothed GLM partial residual plot is a simplified version of the log ratio of smooths, because it requires only one smoothing operation. This plot is a refinement of a diagnostic plot described in TGF. They also recommended plotting scaled martingale residuals, but the residuals come from fitting a Cox model *without* X and the scaling consisted of dividing by the overall proportion of events, $n^{-1} \sum N_i$, rather than by $\hat{E}_i$ values. They suggested superimposing an unweighted scatterplot smooth.

There is also a close relationship between the GLM style residuals and the method proposed in Hastie and Tibshirani (1990a), who consider the simultaneous estimation of the functional form of each of the covariates in the Cox model. They assume that the hazard is $\lambda_0(t) \exp\{\sum_{j=1}^p f_j(Z_j)\}$ where the f_j are unspecified smooth functions. To estimate them, they maximize the penalized partial likelihood with penalty function $\sum_{j=1}^p \lambda_j \int f_j''(s)^2 ds$, where $\lambda_j \geq 0$ ($j = 1, \dots, p$) are user-specified smoothing parameters. Let $\eta_i = \sum_{j=1}^p f_j(z_{ij})$. Then the partial likelihood derivatives are:

$$\frac{\partial l}{\partial \eta_i} = M_i$$

and

$$\begin{aligned}
 -\frac{\partial^2 l}{\partial \eta_i^2} &= \int_0^\tau p_i[1 - p_i] dN_i(t) \\
 &= E_i - \int_0^\tau p_i^2 dN_i(t) \\
 &\equiv \nu_i,
 \end{aligned}$$

where

$$p_i = \frac{Y_i(t)e^{\eta_i}}{\sum_j Y_j(t)e^{\eta_j}}$$

and $N_i(t) = \sum N_i(t)$. Their iterative algorithm involves repeated application of a weighted cubic smoothing spline to the scaled martingale residual, M_i/ν_i , plus the current estimated functional form $\hat{f}_j$ with weights given by ν_i . The first step is effectively the weighted smooth of the GLM partial residual plot proposed here, but with a particular smoother and with weights that differ from $\hat{E}_i$ by a factor of $O(n^{-1})$.

Iteration of our procedure should lead to approximately the same solution as Hastie and Tibshirani's method. In fact, iteration may not be required; examples later in this paper show that our "one step" approach is often accurate. An advantage of our approach is that special regression software is not necessary. Related techniques for estimating smooth f_j terms include the local full likelihood approach of Gentleman and Crowley (1991) and the penalized regression splines with a moderate number of knots from Gray (1992). These also require special software, since the optimization is done within the kernel of the partial likelihood.

3. Simulation for the Cox Model

The semiparametric Cox model for censored survival data (Cox, 1972) is a popular special case of the multiplicative intensity model. This section uses Monte Carlo simulations to explore the behavior of the diagnostic procedures for Cox regression. The proposed techniques are compared to the earlier recommendations of TGF, and their robustness under varying degrees of model misspecification is assessed.

We consider three models with two predictors. The failure time is exponential with hazard rate $\lambda_0 \exp\{f_1(Z_1) + f_2(Z_2)\}$. The first model is log-linear; $f_1(Z_1) = 1.7Z_1$ and $f_2(Z_2) = \beta_2 Z_2$ with $\beta_2 = 1.7$ or 3.4 . The second model is log-linear in only the first predictor; $f_1(Z_1) = 1.7Z_1$ and $f_2(Z_2) = .63(Z_2 - .6)^3$. The third model has extreme nonlinearity in both predictors: $f_1(Z_1) = 7.2Z_1 - \log\{6,000 + \exp(7.2Z_1)\}$ and $f_2(Z_2) = \exp\{(Z_2 - 1)^2\}$. The covariates are bivariate normal with common mean $\mu = 1.5$, common standard deviation .3, and one of four possible correlations: .0, .6, .75, or .90. Most covariates lie in $[1, 2]$ and the functions are chosen so that the risk of failure for $Z = 2$ is $\exp(1.7)$ times that for $Z = 1$. We set $\lambda_0 = \exp\{-f_1(\mu) - f_2(\mu)\}$. Censoring comes from an independent uniform on $[0, c]$ with c chosen to give an overall censoring rate of roughly 27%.

Each simulated data set consists of $n = 133$ observations and 1,000 data sets are generated for each model and correlation combination, all with the same set of predictors. A Cox model with hazard rate log-linear in Z_1 and Z_2 is fit to each data set and both new diagnostic techniques, the weighted smooth of GLM partial residuals and the log of the ratio of smooths of N_i and $\hat{E}_i$ versus Z_{ij} , $j = 1, 2$, are computed. For the first two models, the two proposals of TGF are also implemented. Specifically to reveal the functional form for Z_j , $j = 1, 2$, we fit a single predictor log-linear Cox model which does not involve Z_j . The first TGF proposal entails plotting N_i against Z_{ij} and smoothing, plotting the estimated expected values from the fit against Z_{ij} and smoothing and taking the log of the ratio of the smooths. Their second proposal is a smooth of the martingale residuals divided by the proportion of failures in the data set and plotted against Z_{ij} .

In all cases the smoother is a natural cubic regression spline with three interior knots at the first three quartiles of Z . This smoother has the advantage of ready availability and quick computation, since it involves standard linear regression on a four degree of freedom natural spline basis. There is a disadvantage: the smooths of N_i or $\hat{E}_i$ may give inappropriate nonpositive values and result in an undefined value for the log of the ratio of the smooths. In the case of a monotone relation between hazard and covariate, this problem typically occurs only at the edge of the scatter plot where observed and/or expected counts are small. It can be ameliorated, but not eliminated, by discarding

all noninformative data points, cases of censoring before the first failure. This is done in these simulations.

For summarizing the simulation results, the fitted smooths for each diagnostic plot are computed at 41 points equally spaced in $[1, 2]$; i.e., 1.00, 1.025, $\dots$, 2.00. This interval covers 90% of the data and nearly eliminates the problem of undefined values for the ratio of smooths. Because the Cox model is semiparametric, the intercept in a diagnostic plot is not identifiable. For comparability among simulations, each smooth is adjusted by an additive constant so that it passes through the point (1.5, 2). The simulations are summarized by the 41 point-wise medians of the 1,000 adjusted smooths. Variability across data sets is shown by nonparametric 80% prediction intervals; the 10th and 90th quantiles of the diagnostic plots are computed at $Z = 1.00, 1.25, 1.75$, and 2.00.

The first three figures give results for the log-linear model. Figure 1 shows GLM partial residual plots for two of the randomly generated data sets. As is typical for censored failure time data, it is difficult to see any structure, because the partial residuals are heavily skewed with long right-hand tails. In Figure 2, the plots are rescaled to show the diagnostic curves. Weighted and unweighted smooths of partial residuals are compared. The weighted scatter plot smooths are roughly linear although the smoother is overfitting, as expected. The instability of the unweighted smooths is evident. Figure 3 compares the four diagnostic techniques in estimating $f_2(Z_2)$ when Z_1 and Z_2 are independent. In the top panel $f_2(Z_2) = 1.7Z_2$, and in the bottom, $f_2(Z_2) = 3.4Z_2$. The four techniques are fairly similar, but the superiority of the two new techniques is evident. Their median smooths are virtually indistinguishable from the true f_2 . The ratio of the log of the smooths plot has slightly smaller variability than the GLM partial residuals plot. We conjecture that this is due to the fact that the log ratio of smooths is based on the difference of two positively correlated quantities, thus decreasing the variance. As theory predicts, the TGF proposals perform quite well when Z_1 and Z_2 are uncorrelated, unless the impact of Z_2 on the hazard is large (bottom graphs, Figure 3), but with dependency between the Z covariates, their performance markedly deteriorates (cf. Figure 4).

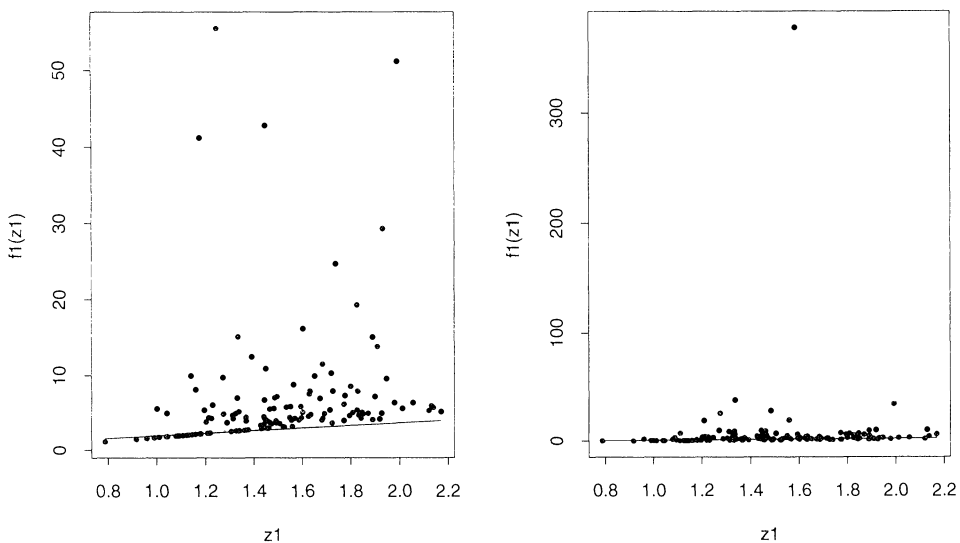


Figure 1. Plot of GLM partial residuals from two simulated data sets. The line gives the true functional form.

Figure 4 compares the various methods in estimating f_2 for model 2, where the true functional form is nonlinear, for various correlations between the predictors ($\rho = .0, .60, .90$). The two TGF proposed plots behave similarly. Their median smooth provides a reasonable fit to the underlying f_2 when the predictors are uncorrelated, but with increasing correlation, their median smooth flattens and departs substantially from f_2 . The new techniques track f_2 very well irrespective of covariate correlation. The log ratio of smooths is marginally superior to the weighted GLM partial residuals. Its median smooth fits slightly better, and its variability is somewhat less.

The simulations for model 3 explore the performance of the new techniques when the postulated model has incorrect functional form for both covariates. Figure 5 shows the results for $\log(\text{smooth } N / \text{smooth } E)$ diagnostic plots. With independent predictors, the median smooth is virtually identical to the true functional form. The performance deteriorates as the correlation between the predictors increases. The diagnostic curves for f_1 and f_2 become more similar as Z_1 and Z_2 become

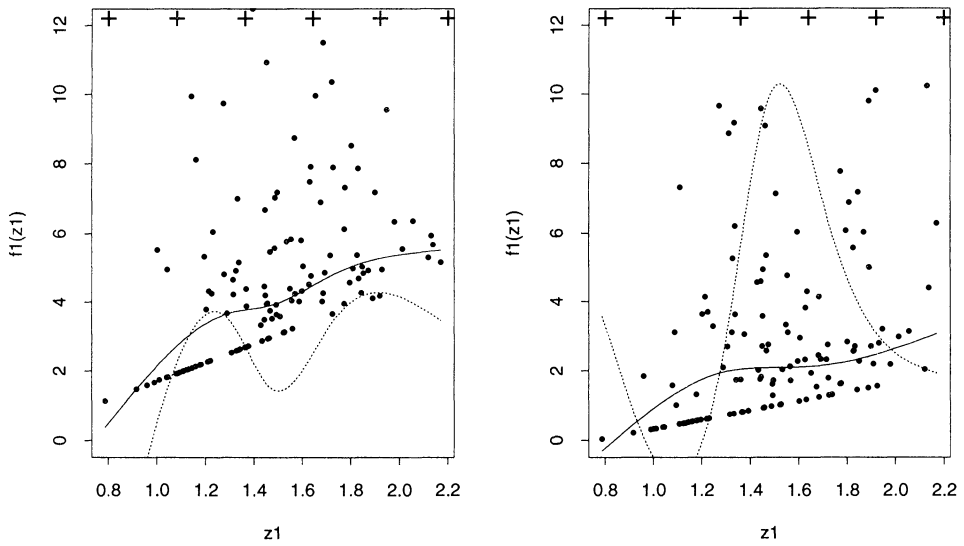


Figure 2. Scatter plot smooth (natural spline with three interior knots) superimposed on the residuals of Figure 1. The solid line shows a weighted smooth, and the dotted line shows an unweighted smooth. +, presence of points off the scale.

more highly correlated. Since the true f_1 and f_2 are quite different, the bias in the diagnostic curves must increase. Curves for the weighted partial residuals (not shown) behave similarly.

This problem can be nearly eliminated by borrowing a technique for partial residual plots developed in ordinary least-squares regression. To deal with highly correlated predictors, Mallows (1986) suggested the *augmented partial residual plot*, in which the regressors were quadratic polynomials in the covariates rather than linear terms and in which the partial residual plot was of

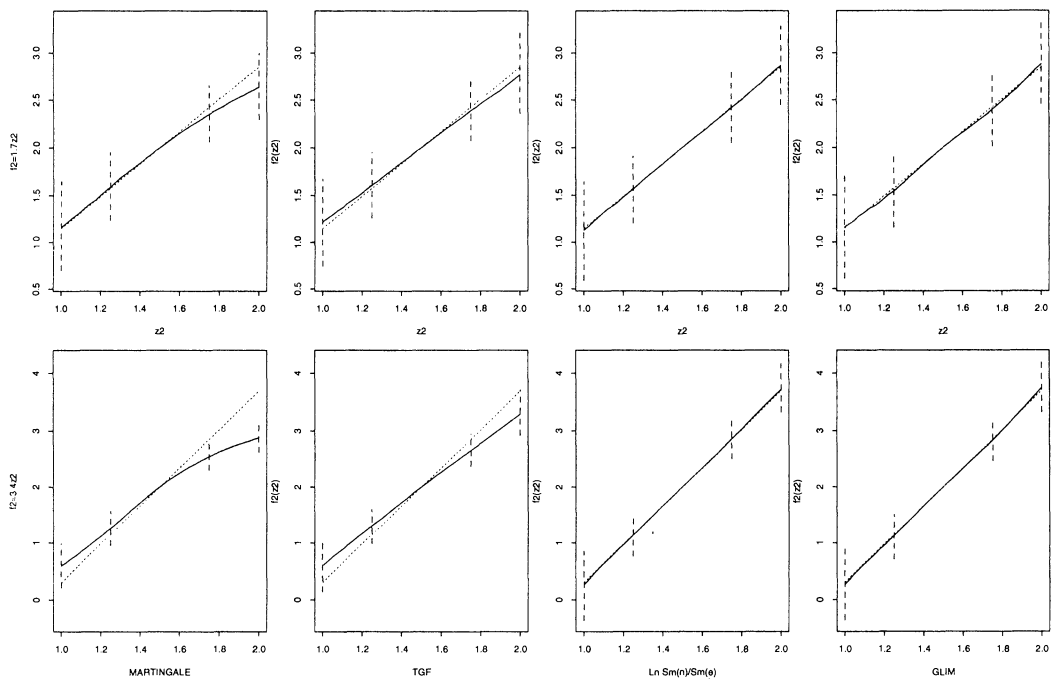


Figure 3. Summary of the four diagnostic techniques to estimate $f_2(Z_2)$ applied to data from a log-linear model with uncorrelated predictors. In the top panel, Z_2 has a moderate impact on hazard ($\beta_2 = 1.7$), and in the bottom panel Z_2 has a very large impact ($\beta_2 = 3.4$). The first two columns show the TGF proposals, and the last two give the new techniques. The solid line is the median of 1,000 smooths, and the dotted line is $f_2(Z_2)$. The vertical dashed lines go from the 10th to the 90th quantiles of the set of smooths.

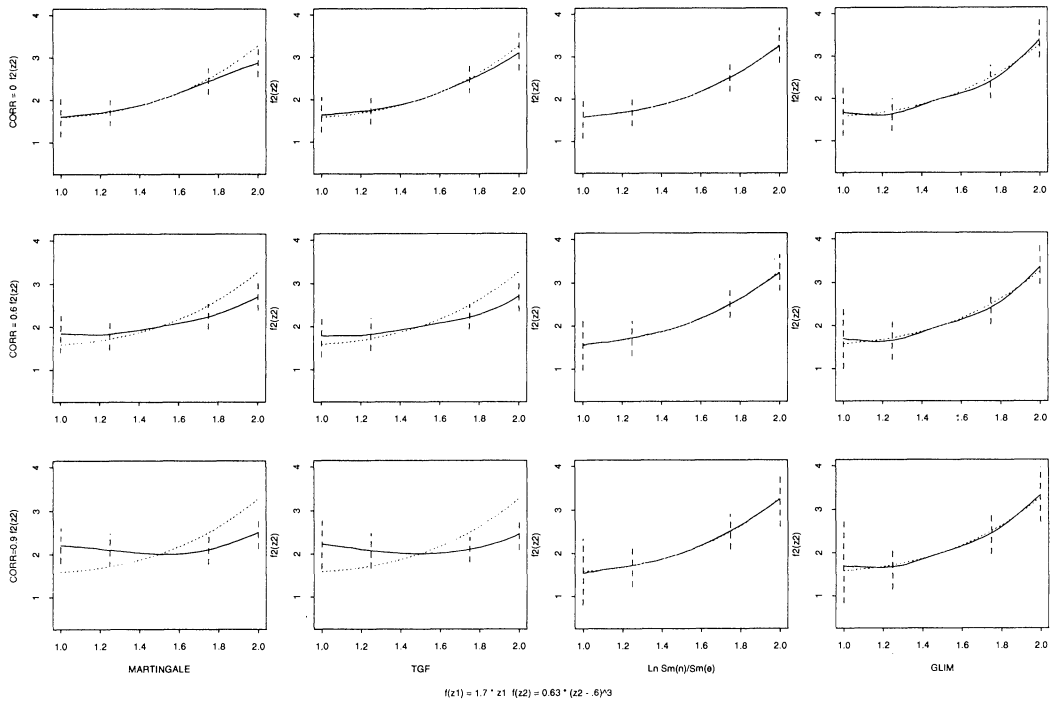


Figure 4. Summary of the four diagnostic techniques to estimate $f_2(Z_2)$ applied to data from a model log-linear in Z_1 , but not in Z_2 . The estimated model was log-linear in both predictors. The correlation between the predictors was .0 in the top row, .6 in the middle row, and .9 in the bottom row. Figure symbols are the same as those in Figure 3.

(residual + $\hat{\gamma}_1 Z + \hat{\gamma}_2 Z^2$) versus covariate Z . That approach seems to work well for the Cox model, too. Figure 6 shows the results for the log(smooth N /smooth E) diagnostic plot when Z_1 and Z_2 have correlation .9. The bias is nearly eliminated with only a slight increase in variability. This substantial improvement occurs despite the fact that the true model is not log-quadratic.

4. Applications

These diagnostic procedures were applied to two survival data sets. The first is the primary biliary cirrhosis data from Appendix D of Fleming and Harrington (1991). Primary biliary cirrhosis is a progressive, ultimately fatal liver disease, presently incurable, with liver transplantation as the only life-saving treatment. A survival model was developed using the 312 patients in this data set. The model was a five-variable Cox model with age, edema severity, prothrombin time, albumin, and bilirubin. Bilirubin was the most important predictor. To assess its functional form, two models were considered. In the first, bilirubin appeared in its original scale and in the second, the natural logarithm was used. The remaining four variables had the same functional forms in both models; albumin and prothrombin time were in the log scale, edema severity had three ordered categories, and age was in years. Figure 7 presents the GLM partial residual plots for bilirubin for the two models. The left panel shows the functional form for bilirubin in the original scale and the right shows the log transform. The smoother used was loess (Cleveland, 1979), a locally weighted least-squares regression with a span of 60% of the data, giving roughly four equivalent degrees of freedom. For comparison to the previous TGF proposal, we include the smoothed and scaled martingale residual from the four covariate model without bilirubin.

The two smooths were translated to pass through 0 at a common point near the median of bilirubin, 1.35. To assess goodness-of-fit, the estimated function from each model, $\hat{\gamma}_1 \times \text{bilirubin}$ for model 1 and $\hat{\gamma}_2 \times \log(\text{bilirubin})$ for model 2, is superimposed. With model 1, the two residual smooths are similar to each other and quite distinct from the estimated line. This implies that the linear functional form is wrong. The shape of the residual smooths suggests that a concave transform would fit better. The two smooths are less similar for model 2. The weighted GLM partial residual smooth improves over the scaled martingale residual and is quite close to the estimated functional

LINEAR MODEL FIT

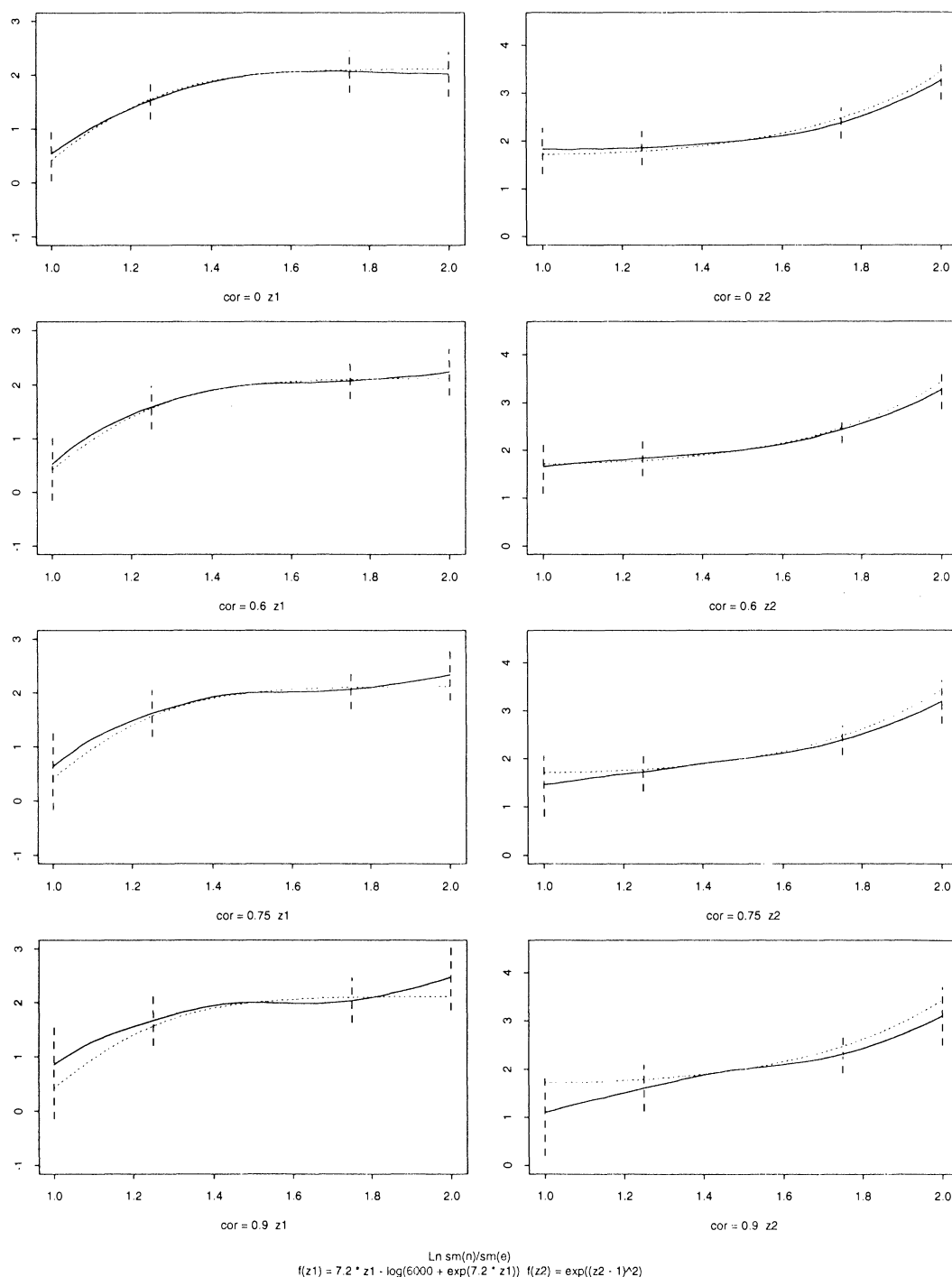


Figure 5. Summary of the log of the ratio of smooths for data from a non-log-linear model in both predictors where the estimated model was log-linear in both predictors. The left column shows the estimator for $f_1(Z_1)$, and the right column shows that for $f_2(Z_2)$. The rows reflect different correlations between the predictors: (.0, .60, .75, and .90). Figure symbols are the same as those in Figure 3.

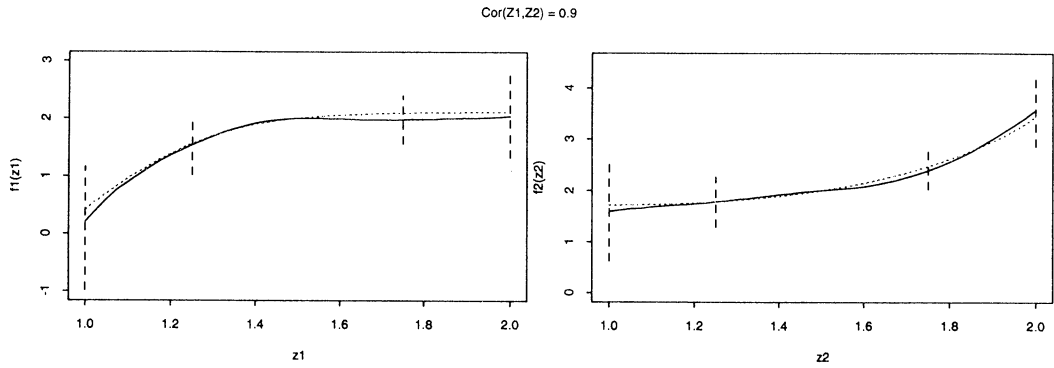


Figure 6. Summary of the log of the ratio of smooths for the same data as in the bottom row of Figure 5. The estimated model was log-quadratic in both predictors. Figure symbols are the same as those in Figure 3.

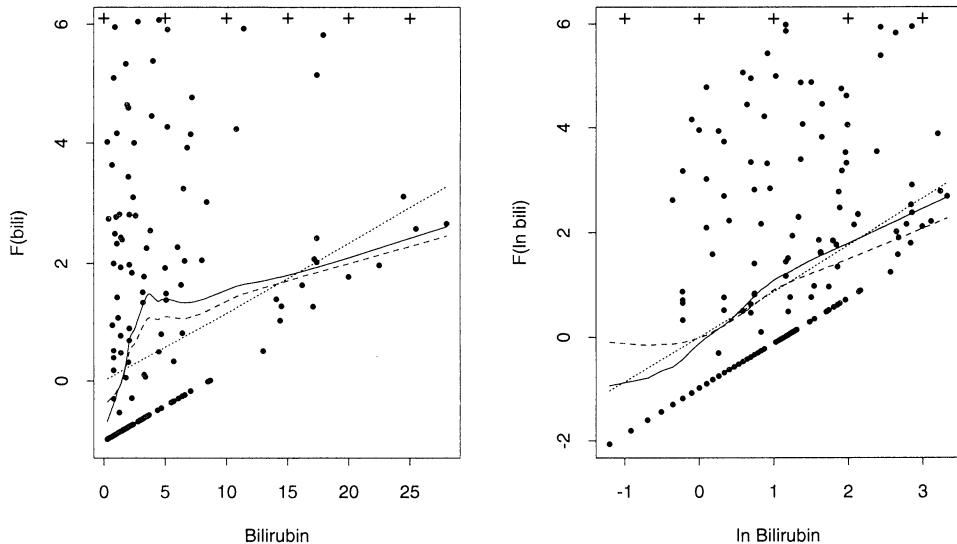


Figure 7. GLM partial residual plots for the covariate bilirubin for the PBC model. The left plot comes from a model linear in bilirubin, and the right comes from a model with log bilirubin. The solid line gives the weighted smooth GLM partial residual, the dashed line gives the scaled smoothed martingale residual from a model without bilirubin, and the dotted line gives the estimate based on the model. The last two lines are adjusted to have the same median as the first.

form. The plots show clearly that bilirubin in the log scale gives a better fit; Fleming and Harrington (1991, p. 180) reached the same conclusion using only scaled martingale residual plots.

Arjas martingale plots were also done for these data. Five groups were formed, using {1.2, 2.5, 4.2, 8.5} as cut points for bilirubin, to make groups with roughly equal numbers of deaths. Figure 8, left panel, shows the results from fitting the model without bilirubin, which was Arjas' original formulation. The five groups are nearly monotonically ordered and well separated. The right panel gives the curves resulting from a model including log bilirubin. The curves are much closer together, showing better fit for this model. However, both the GLM partial residual plot (Fig. 7, right panel) and the Arjas plot suggest there may still be some lack of fit for the log model. To assess this, we used the hypothesis-testing procedure for functional form described in Andersen et al. (1993, §VII, 3.2) and recommended by a referee. We added four indicator variables for the five groups to model 2. The change in log-likelihood was small (from -540.30 to -536.46 on four degrees of freedom) and not statistically significant at conventional levels, nor were any of the group indicators. We conclude that most of the impact of bilirubin is captured by a log transformation.

The second example involved the leukemia data set, presented in Kalbfleisch and Prentice (1980), and used by Hastie and Tibshirani (1990a) to illustrate their methodology. These data came from a study designed to examine the influence of genetic and viral factors on the development of

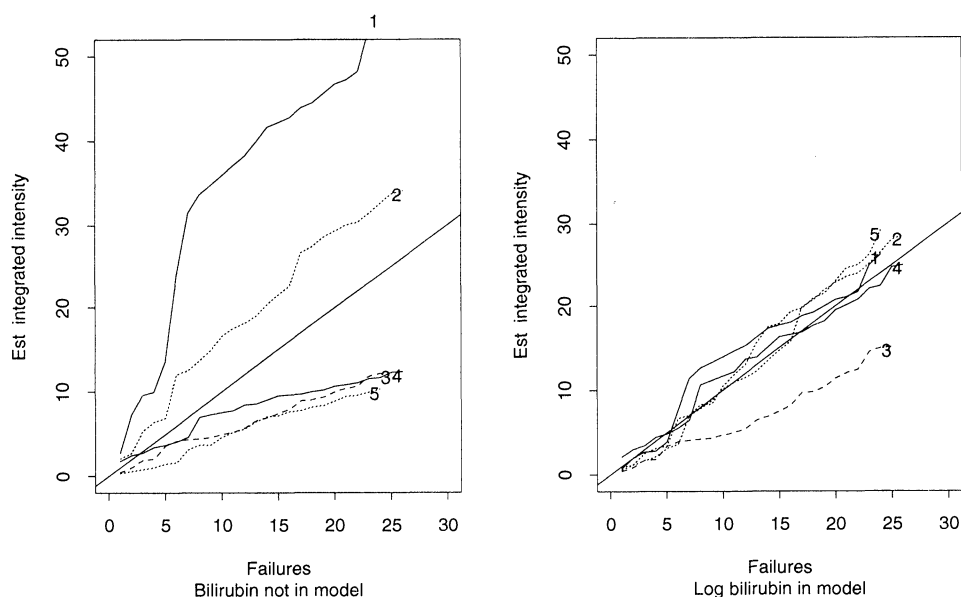


Figure 8. Arjas plots for the PBC data of Figure 7 using the same two models. The five lines are plots of the cumulative estimated expected counts versus observed counts of failures for five groups of participants, categorized by level of bilirubin.

spontaneous leukemia in mice. We used the same subset as Hastie and Tibshirani, 175 mice with complete data and considered their model (10), a Cox model involving two continuous predictors, antiviral antibody level (% gp 70 precipitate) and virus level (PFU/ml), and three binary covariates: coat color, gender, and MHC phenotype. Figure 9 compares residual plots for the continuous predictors from a model log-linear in the five predictors. The smoother used was a smoothing spline with four equivalent degrees of freedom, similar to the smoother used by Hastie and Tibshirani. For each covariate, the log ratio of smooths, the weighted GLM partial residual plot, and the smoothed and scaled martingale residuals from a model omitting that covariate are shown. The three curves give roughly the same picture for each variable, although the scaled martingale residual which leaves out the variable in question offers some distortion. The two new techniques are similar to each other.

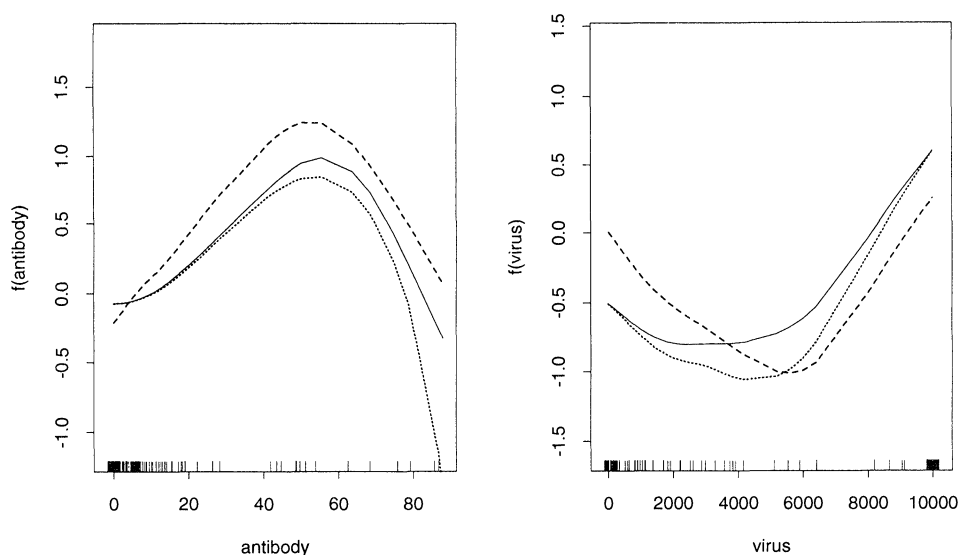


Figure 9. Smoothed residual plots for the leukemia data. The solid line gives the weighted smooth GLM partial residual, the dotted line gives the log ratio of smooths, and the dashed line gives the scaled smoothed martingale residual from a model omitting the covariate of the plot.

A comparison with Figure 1 in Hastie and Tibshirani's paper shows that these methods give basically the same functional form as was estimated by Hastie and Tibshirani.

5. Discussion

We have presented plotting techniques for diagnosing the functional form of covariates in proportional hazards models. The advantage of our techniques is accessibility. Any statistical software, which fits the Cox model, computes martingale residuals, and provides scatter plot smoothers, can be used. The techniques can be extended. We have focused on time-fixed covariates, but the methods may be extended to time-varying covariates as well. Assume that the time-dependent covariate is piecewise constant over time intervals (such as the treatment covariate in the Stanford heart transplant study). Then all of the development found in Section 2 applies, but treating each of the time intervals as a separate observation.

As a referee has pointed out, the model diagnostics for a particular covariate X involve conditioning on X only, averaging over the other covariates, whereas the estimation procedures for the Cox likelihood condition on all of these. The reason for this averaging is to borrow strength. In very large data sets, particularly if some of the other (i.e., Z) covariates were discrete, one could condition on the value of the Z terms as well, producing a diagnostic plot for X for each of several distinct Z strata. Differences among these plots would suggest interaction between X and some of the Z terms. Thus, the technique can be used to detect covariate interactions, as well as functional form for particular covariates.

The plots have limitations, and further research is needed. Measures of precision (confidence limits and standard error estimates) need to be developed. We conjecture that the methodology for confidence limits for scatter-plot smooths of Schoenfeld residuals (Grambsch and Therneau, 1994) could be applied here, as well; although, at present, it is restricted to the case where the model fit to the data has the correct functional form. The skewness of the martingale residuals, so obvious in Figure 1, has been noted (Therneau et al., 1990; Aalen, 1993), and is a potential drawback, particularly since most readily available scatterplot smoothers assume a symmetric error structure. It is possible that designing special smoothers for martingale residuals or smoothing the data in a transformed scale (such as log) and then back-transforming would improve the plots, at the expense of additional complexity of implementation.

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RÉSUMÉ

Des graphiques basés sur les résidus d'un modèle à risque proportionnels peuvent être utilisés pour révéler la forme fonctionnelle correcte pour les covariables dans le modèle. A cet égard, un graphique lissé des résidus de la martingale a été suggéré par Therneau, Grambsch et Fleming (1990, *Biometrika* **77**, 147–160). Cependant, l'indépendance entre les covariables était nécessaire pour assurer la consistance de cette méthode. Les auteurs ont également noté que le graphique pouvait être biaisé en cas d'effets importants des covariables. Dans cet article, on introduit deux améliorations qui maîtrisent ces difficultés. La première est basée sur un rapport de graphiques lissés, où le numérateur correspond au lissage du nombre observé par rapport à la covariable, et le dénominateur correspond au lissage du nombre attendu. Cette technique est proche de la méthode graphique d'adéquation d'Arjas (1988, *Journal of the American Statistical Association* **83**, 204–212). La seconde consiste à lisser les résidus des martingales divisé par le nombre attendu, en prenant ce nombre comme facteur de pondération. Cette dernière technique est proche de la méthode graphique des résidus GLM partiels, ainsi que des méthodes itératives de Hastie et Tibshirani (1990, *Biometrics* **46**, 1005–1016) et Gentleman et Crowley (1991, *Biometrics* **47**, 1283–1296). Des applications aux données de survie sont présentées.

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APPENDIX: ASYMPTOTIC BEHAVIOR OF $\hat{E}$ UNDER THE MISSPECIFIED MODEL

Recall that $\hat{E} = \int_0^\tau Y(t) \exp(\hat{\gamma}X + \hat{\beta}'Z) d\hat{\Lambda}_0(t)$, where one has fit a model assuming that the hazard function is log-linear in X and Z to a data set derived from a model with hazard function $h(X) \exp(\beta'Z) \lambda_0(t)$, not log-linear in X . We treat the problem as a model with an “omitted” covariate, modifying the approach in Appendix 1 of TGF, because the omitted covariate, $h(X) \exp(-\gamma^*X)$, is functionally dependent on one of the model covariates. Let

$$\begin{aligned} \bar{h}(t) &= \frac{E\{h(X) \exp(\beta'Z) | Y(t) = 1\}}{E\{\exp(\gamma^*X) \exp(\beta^{*'}Z) | Y(t) = 1\}} \\ &= \frac{E\{Y(t) h(X) \exp(\beta'Z)\}}{E\{Y(t) \exp(\gamma^*X) \exp(\beta^{*'}Z)\}}. \end{aligned}$$

Then we decompose $\hat{E}$:

$$\int_0^\tau Y(t) \exp(\hat{\gamma}X + \hat{\beta}'Z) d\hat{\Lambda}_0(t) = \int_0^\tau Y(t) \exp(\gamma^*X + \beta^{*'}Z) \bar{h}(t) \lambda_0(t) dt + o_p(1), \quad (\text{A1})$$

since the left-hand side is a consistent estimator of the first term of the right-hand side, where this first term will be denoted by E^* . The proof of (A1) follows from the theory of maximum likelihood estimation with misspecified models (Bickel et al., 1993), yielding the uniform consistency of $\hat{\gamma}$ and $\hat{\beta}$ for γ^* and β^* , respectively; from the uniform strong consistency of $\frac{1}{n} \sum M_i(s)$ and $\frac{1}{n} \sum Y_i(s)$ which yields the uniform consistency of $\hat{\Lambda}$ for $\int \bar{h}(t) \lambda_0(t) dt$; and from a version of the Helly–Bray lemma:

$$\|(x_n, y_n) - (x, y)\| \rightarrow 0 \Rightarrow \left\| \int x_n dy_n - \int x dy \right\| \rightarrow 0$$

by Andersen et al. (1993, p. 113).

We wish to formulate large sample approximations for $E(\hat{E}|X)$. Since $\hat{E} = E^* + o_p(1)$, it is of interest to consider approximations for $E(E^*|X)$. Let $\bar{h}(X) =$

$$\frac{E\left\{\int_0^\tau Y(t)\exp(\gamma^*X + \beta^{*'}Z)\bar{h}(t)\lambda_0(t) dt|X\right\}}{E\left\{\int_0^\tau Y(t)\exp(\gamma^*X + \beta^{*'}Z)\lambda_0(t) dt|X\right\}}$$

and $\bar{h} = E\{\bar{h}(X)\}$.

Then

$$\begin{aligned} E^* &= \int_0^\tau Y(t)\exp\int_0^\tau (\gamma^*X + \beta^{*'}Z)\bar{h}(t)\lambda_0(t) dt \\ &= \bar{h} \int_0^\tau Y(t)\exp(\gamma^*X + \beta^{*'}Z)\lambda_0(t) dt + \bar{h} \int_0^\tau Y(t)\exp(\gamma^*X)\{\exp(\beta^{*'}Z) \\ &\quad - \exp(\beta^{*'}Z)\}\lambda_0(t) dt + \{\bar{h}(X) - \bar{h}\} \int_0^\tau Y(t)\exp(\gamma^*X + \beta^{*'}Z)\lambda_0(t) dt \\ &\quad + \int_0^\tau Y(t)\exp(\gamma^*X + \beta^{*'}Z)\{\bar{h}(t) - \bar{h}(X)\}\lambda_0(t) dt. \end{aligned} \quad (A2)$$

The fourth summand in (A2) has conditional mean 0, given X , by definition. Thus

$$E(E^*|X) = \bar{h}\exp(\gamma^*X)E\left\{\int_0^\tau Y(t)\exp(\beta^{*'}Z)\lambda_0(t) dt|X\right\} + R_1 + R_2.$$

The two remainder terms are small relative to the leading term when the log-linear assumption is a good approximation to $h(x)$, so that $\bar{h}(t) \approx 1$. In fact, there are circumstances under which they are negligible even when the log-linear assumption is a poor guess. The first remainder term is sensitive to the difference between β and β^* and is 0 when X is the only covariate or when X and Z are independent not only at the start of the study, but also among those still at risk at all event times, $0 \leq t \leq \tau$. In that case, $\beta = \beta^*$. The remainder term R_2 is 0 if $\bar{h}(t)$ is a constant function, so that $\bar{h}(x) \equiv \bar{h} = h(t) = c$. This occurs in Poisson regression when every subject remains at risk throughout the experiment, so that $Y(t) = 1$, $0 \leq t < \tau$.